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1 March 1960

MILITARY SPECIFICATION

ADAPTERS, COMPENSATING, MS28107, MA-1 COMPASS SYSTEM

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers design and performance requirements for the procurement of various types of Compensating Adapters which are utilized with the Type MA-1 Compass System.

1.2 Classification - The Compensating Adapters shall be of the types listed in 6.8.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-M-3171	Magnesium Alloy; Processes for Corrosion Protection of
MIL-C-4150	Case, Carrying, Water-Vaporproof
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for

FSC 6605

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Military (Continued)

MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-M-7793	Meter, Time Totalizing
MIL-C-17858	Compass System; Magnetic Slaved, Directional Gyro Stabilized, Type MA-1
MIL-T-18303	Test Procedures: Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-19576	Transmitter; Remote Compass, Thin Wing Type ML-1 (Unstabilized)
MIL-S-20708	Synchros, 60 and 400 Cycle, General Specification for
MIL-C-26482	Connectors, Electric, Circular, Miniature, Quick Disconnect

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of

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Military (Continued)

MS17321	Meter, Time Totalizing, 28 VDC, Miniature Digital
MS28107	Adapter, Compensating MA-1 Compass

2.2 Availability of documents - When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 1051, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

* 3.2 Parts and materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(a) Parts and materials requirements shall conform to MIL-E-5400.

(b) Nonrepairable subassemblies, as outlined in MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable (see 6.4).

(c) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

* 3.2.1 Nonstandard parts and materials approval - Approval for the use of nonstandard parts and materials other than microelectronic devices shall be obtained as outlined in MIL-E-5400.

* 3.3 Design and construction - The adapter shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein. The adapter shall be capable of compensating the heading signal output of the Type MA-1 Compass System conforming to MIL-C-17858, by correcting the magnetic deviation errors. Provision

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shall be made for compensation adjustments at 24 equally spaced headings including the cardinal points. A minimum of ±3 degrees and a maximum of ±7 degrees compensation shall be available at each compensation point. The design shall be such that no physical damage to the compensator will result from any combination of adjustment settings. The compensation gradient between adjacent taps shall describe a smooth linear curve within the limits specified in 4.7. The adapters shall contain synchro outputs as indicated in 6.8. The design of the adapter shall be reasonably simple to permit overhaul or repair with a minimum of special tools and fixtures. The instrument shall be constructed to withstand the normal strains, jars, vibrations and such other conditions as are incident to shipping, storage, installation and service without undue failures.

3.3.1 Total weight - The total weight of the adapter, including the mounting rack, excluding cables, shall be a minimum consistent with good design and shall not exceed 9.0 pounds.

* 3.3.2 Reliability -

* 3.3.2.1 Operational stability - The adapter shall operate with satisfactory performance, continuously or intermittently for a period of at least 500 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

* 3.3.2.2 Longevity - The adapter shall have total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

* 3.3.2.3 Reliability in mean time between failure (MTBF) - The adapter shall have 1000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.4 Time totalizing meter -The adapter shall include within its case a time totalizing meter conforming to MIL-M-7793 and MS17321 (10,000 hour range).

3.3.3 Cabling and connections -

* 3.3.3.1 Cables and connectors - The equipment shall provide for the use of cables and connectors in accordance with MIL-E-5400.

* 3.3.3.2 Interconnection cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated

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as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

* 3.3.4 Interchangeability of reordered equipment - For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

* 3.3.5 Interference control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of MIL-I-6181.

* 3.3.6 Maintenance provisions and field testing - Provisions for maintenance shall be as specified in MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

* 3.3.7 Nameplates - Nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

* 3.3.8 Standard conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25 ±5° C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ±1.0 VAC and 27.5 ±0.5 VDC

* 3.3.9 Service conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

* 3.3.9.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves II and IV of MIL-E-5400.

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3.3.10 Warm-up time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed 10 minutes unless otherwise stated.

3.3.11 Primary input power requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(a) AC Power (Single Phase) 115 V, Category "C", 40 VA at 0.65 power factor

(b) DC Power 28 V (Grounded Negative), Category "C", 15 Watts

* 3.3.11.1 Degraded performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

* 3.3.12 Standby provisions - The equipment shall provide for a standby position whereby rated voltage shall be applied to all tube filaments. Except for the associated filament circuits, voltages shall not be applied to other parts of the equipment.

* 3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.5 Detail requirements -

3.5.1 Compensator disconnect - Means shall be provided to remove all compensation to permit calibration of the compass transmitter on installation. The disconnect switch shall be protected against accidental changes to the adjustments but accessible and adjacent to the compensator adjustments.

3.5.2 Compensation identification - Legible and durable markings shall be provided to identify the compensation adjustment points.

3.5.3 Case - The outline dimensions of the case shall conform to MS28107. The case shall be finished with a durable, lusterless, black finish, Color No. 37038 of FED-STD-595.

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3.5.3.1 Hermetic sealing - If internal subassemblies of the adapter case are to be hermetically sealed, the subassembly cases shall be so constructed that they may be opened, the mechanisms removed and replaced and the cases reclosed and resealed, at least three times. This shall be possible without the use of any special tool, jig or fixture, unless such device is specifically approved by the procuring activity. The sealing of the cases shall not be dependent upon any material which will be adversely affected by any environment to which the subassembly cases may be subjected in normal use in military aircraft.

3.5.3.2 Filling medium - Hermetically sealed subassembly cases shall be filled to an absolute pressure of 1 ±0.1 atmosphere with an inert gas or a mixture of inert gases. If the leakage test of 4.8.4 is to be performed with a mass spectrometer, the mixture shall contain at least 98 percent helium. The gas or mixture of gases shall be free of dust particles and shall contain not more than 0.006 milligram of water vapor per liter (dew point -65° C) at the filling pressure.

3.5.4 Vibration isolation - The adapter shall be provided with vibration isolators incorporated in the mounting rack.

3.5.5 Output synchros - The nominal electrical characteristics of the output synchros shall be as specified in Table I. Electrical characteristics are specified as nominal values at 25° C and shall be tested as specified in MIL-S-20708.

TABLE I

Input Voltage	26 V - 400 cps
Input Current	0.171 Amp
Input Power	0.7 Watt
Input Impedance	24 + j 150 Ohms
Output Voltage Max. (Line to Line)	11.8 Volts
DC Rotor Resistance	10.3 Ohms
DC Stator Resistance (Line to Line)	3.85 Ohms

3.5.5.1 Clutched synchros - When tuned with the proper phasing capacitor, the clutched synchro shall supply to a 10,000 ohm resistive load a voltage having a minimum gradient of 0.2 volt per degree and a change in phase angle referred to input voltage of less than 10 degrees.

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3.5.6 Undervoltage protection - The adapter shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within its correct range.

3.5.7 Wiring - The external wiring shall be in accordance with Figures 1, 2, 3, 4 and 5. Internal wiring shall be color coded, neat, and accomplished in such a manner that individual wires may be easily traced. Wires shall be secured mechanically to the terminals prior to soldering where practicable. The wiring shall be insulated from the case.

3.5.8 Electrical connector - The electrical connector used shall conform to MIL-C-26482. The type connector used shall be as shown on MS28107. The connector code number markings as specified in MIL-E-5400 shall be legibly and permanently applied either on the connectors or adjacent to each connector on the case of the adapter.

3.5.9 Magnesium alloy parts - Magnesium alloy parts shall be treated in accordance with MIL-M-3171. When abrasion resistance is a factor, an anodic treatment approved by the procuring activity shall be used.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Classification of tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(a) Preproduction Tests

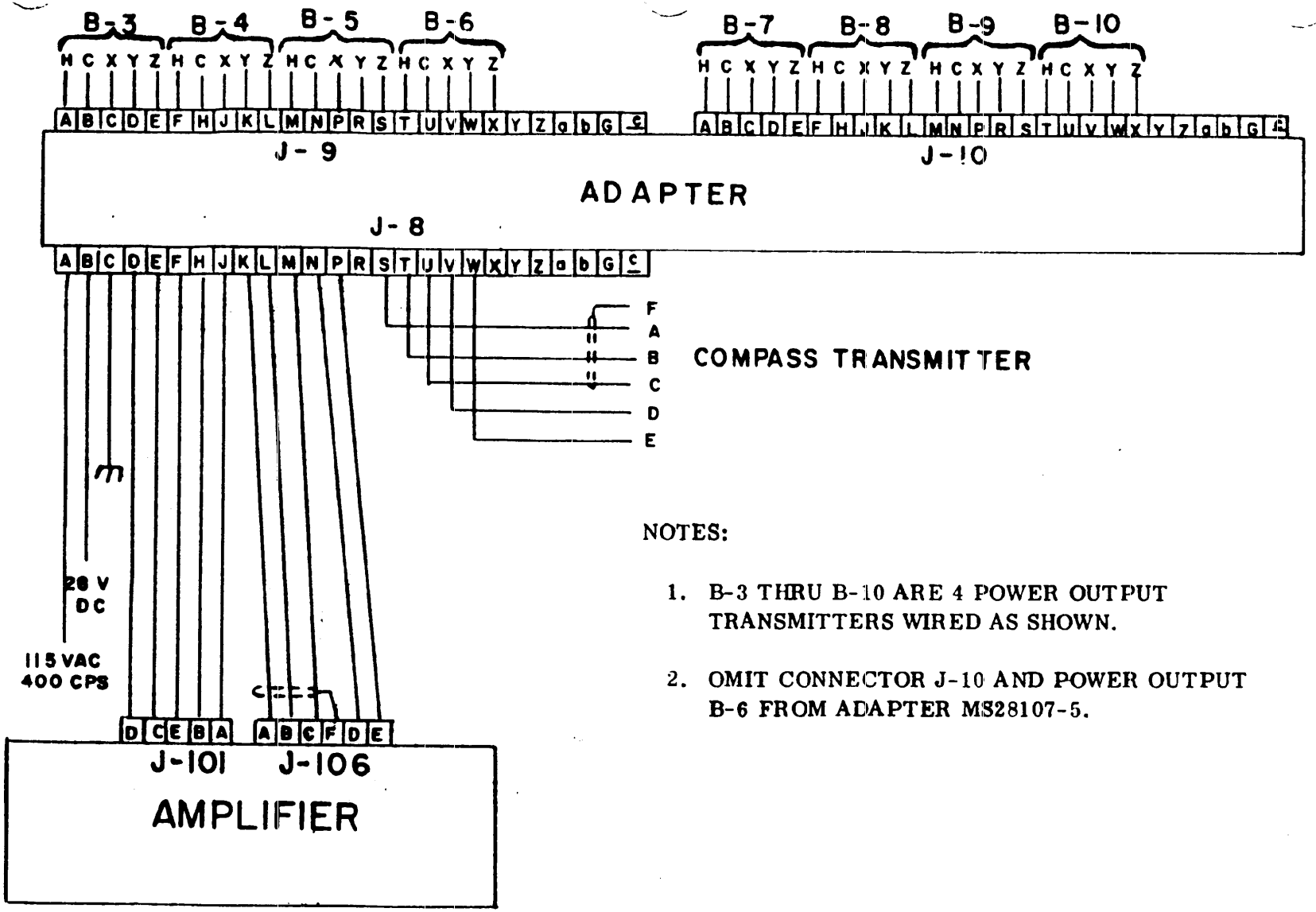
(b) Initial Production Tests

(c) Acceptance Tests

(d) Longevity Test

* 4.2 Preproduction tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility

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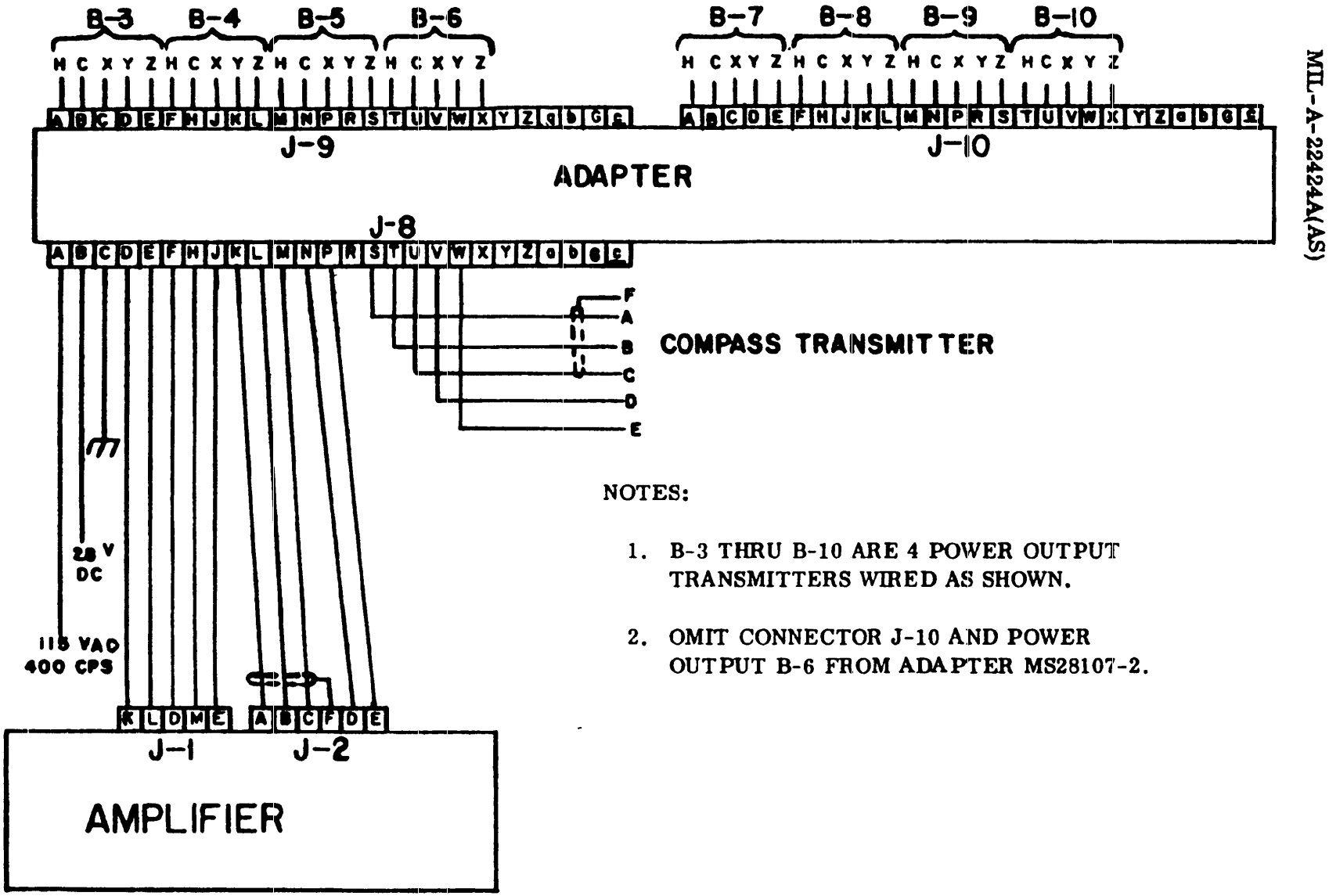


NOTES:

1. B-3 THRU B-10 ARE 4 POWER OUTPUT TRANSMITTERS WIRED AS SHOWN.
2. OMIT CONNECTOR J-10 AND POWER OUTPUT B-6 FROM ADAPTER MS28107-5.

FIGURE 1. EXTERNAL WIRING FOR MS28107-4 AND MS28107-5 COMPENSATING ADAPTERS

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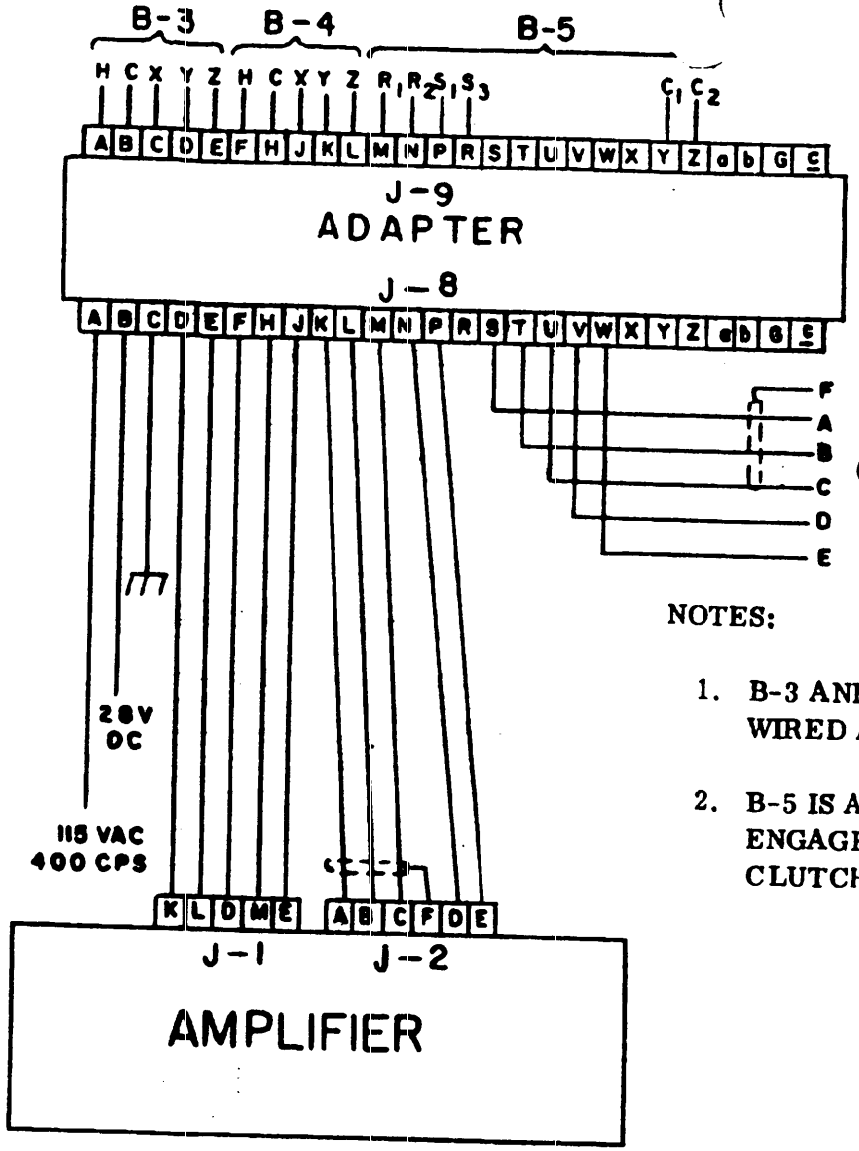


NOTES:

1. B-3 THRU B-10 ARE 4 POWER OUTPUT TRANSMITTERS WIRED AS SHOWN.
2. OMIT CONNECTOR J-10 AND POWER OUTPUT B-6 FROM ADAPTER MS28107-2.

FIGURE 2. EXTERNAL WIRING FOR MS28107-1 AND MS28107-2 COMPENSATING ADAPTERS

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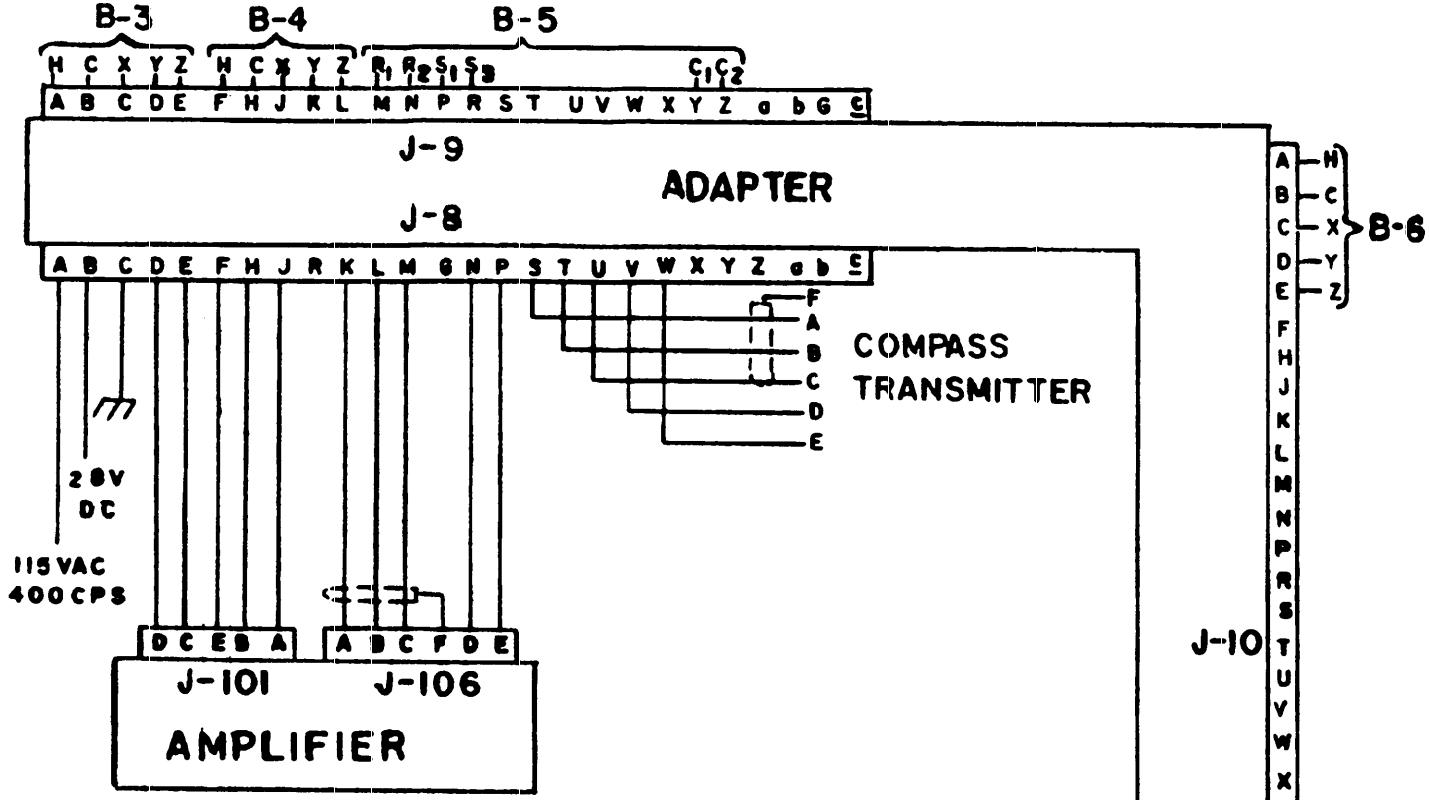


NOTES:

1. B-3 AND B-4 POWER OUTPUT TRANSMITTERS WIRED AS SHOWN.
2. B-5 IS A CLUTCHED SYNCHRO WHICH BECOMES ENGAGED WHEN SIGNAL IS APPLIED TO THE CLUTCH, DESIGNATED AS C_1 AND C_2.

FIGURE 3. EXTERNAL WIRING DIAGRAM FOR MS28107-3 COMPENSATING ADAPTER

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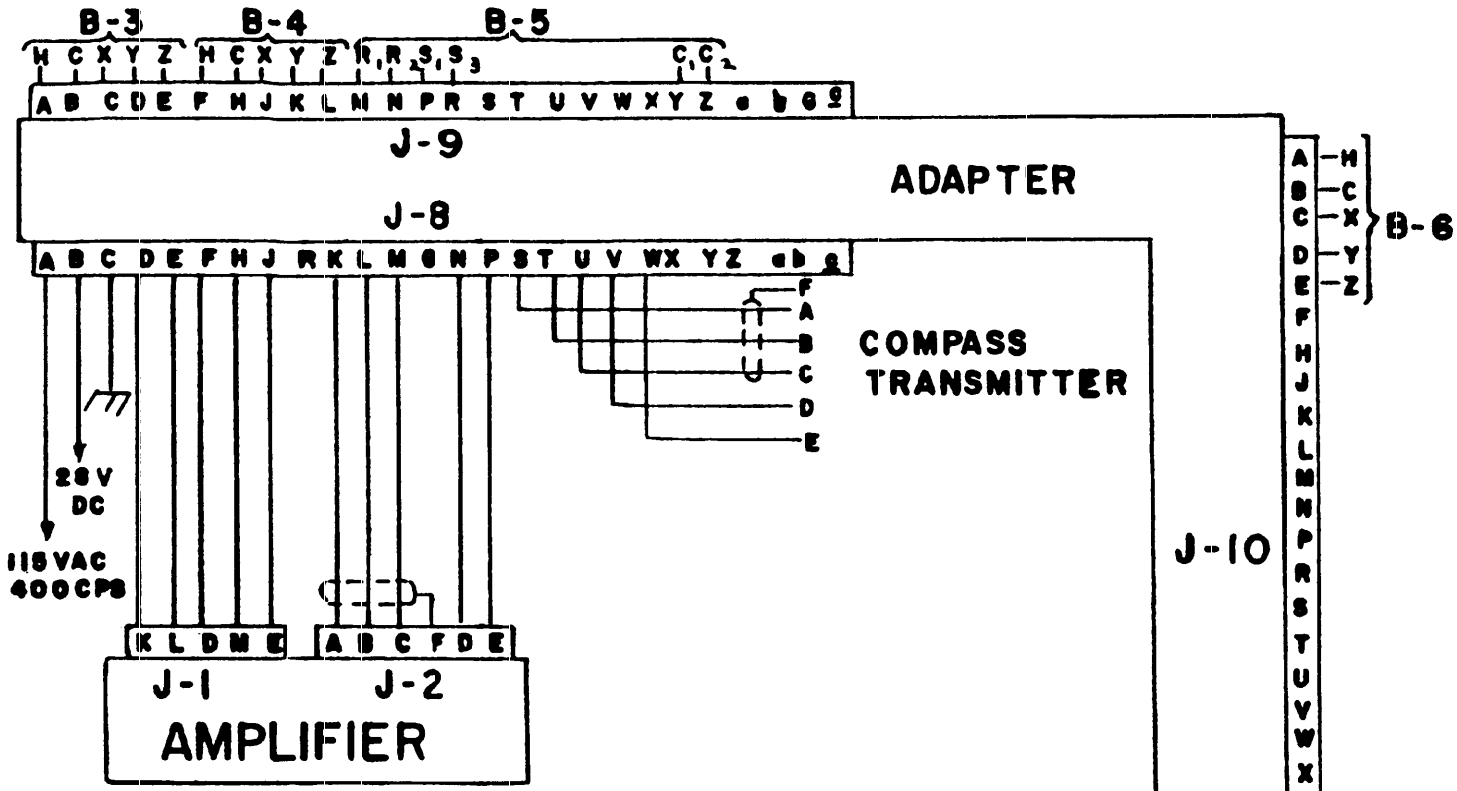


NOTES:

1. B-3, B-4 AND B-6 ARE 4 POWER OUTPUT TRANSMITTERS WIRED AS SHOWN.
2. B-5 IS A CLUTCHED SYNCHRO WHICH BECOMES ENGAGED WHEN SIGNAL IS APPLIED TO THE CLUTCH, DESIGNATED AS C_1 AND C_2.

FIGURE 4. EXTERNAL WIRING FOR MS28107-6 COMPENSATING ADAPTER

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NOTES:

1. B-3, B-4 AND B-6 ARE 4 POWER OUTPUT TRANSMITTERS WIRED AS SHOWN.

2. B-6 IS A CLUTCHED SYNCHRO WHICH BECOMES ENGAGED WHEN SIGNAL IS APPLIED TO THE CLUTCH, DESIGNATED AS C_1 AND C_2.

FIGURE 5. EXTERNAL WIRING FOR MS28107-7 COMPENSATING ADAPTER

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of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

* 4.2.1 Preproduction test data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

* 4.2.2 Scope of tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract, and as a minimum shall consist of all the tests specified herein under test methods.

* 4.2.3 Preproduction approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

* 4.2.4 Production equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

* 4.3 Initial production tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

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* 4.3.1 Scope of tests - This equipment shall be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements and as a minimum shall be subjected to the following tests:

(a) Sampling Tests

(b) Vibration Failure

(c) Radio Noise Interference

(d) Temperature Altitude

(e) Temperature Shock

(f) Impact Shock

(g) Humidity

(h) Salt Spray

(i) Fungus Resistance

* 4.3.2 Accessory material - In addition to the complete equipment submitted for Initial Production Tests, the contractor shall also submit such accessory material and data necessary to test the equipment.

* 4.3.3 Initial production sample approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

* 4.3.4 Reconditioning of initial production test sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

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* 4.4 Acceptance tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports, showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(a) Individual Tests

(b) Sampling Tests

(c) Reliability Assurance Tests

(d) Special Tests

4.4.1 Individual tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(a) Examination of Product	(4.8.1)
(b) Dielectric Strength	(4.8.2)
(c) Scale Error	(4.8.3)
(d) Leakage	(4.8.4)
(e) Compensator Range	(4.8.5)
(f) Servo Response	(4.8.6)

* 4.4.2 Sampling tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

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Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	0 1/
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

1/ If by contract action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(a)	Power Consumption	(4.8.7)
(b)	Voltage and Frequency Variation	(4.8.8)
(c)	Low Temperature Operation	(4.8.9)
(d)	High Temperature Operation	(4.8.10)
(e)	Compensated Scale Error	(4.8.11)

* 4.4.3 Reliability assurance tests - Reliability assurance tests shall be conducted as required by MIL-STD-781. Equipments selected for reliability assurance tests shall first have passed the individual tests. Qualification Phase and Sampling Phase Tests shall be conducted.

* 4.4.3.1 Reliability qualification phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests." The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level F shall be used. The Accept-Reject Criteria for Test Plan IV shall be used.

* 4.4.3.2 Reliability sampling phase tests - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the section entitled "Sampling Phase of Production Reliability Tests." For the Sampling Phase, Test Level F shall be used. The Accept-Reject criteria for Test Plan VI shall be used to determine the length of the tests (until an accept or reject decision is reached).

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* 4.4.3.2.1 Lot size for sampling phase - The equipments constructed during one month shall be one lot.

* 4.4.3.3 Test details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted to and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests. The following shall be considered minimum conditions and shall apply to both the Qualification and Sampling Phases.

* 4.4.3.3.1 Length of heat portion of cycle - After stabilization at the high temperature limit required by the test level, the equipment shall be operated 2-1/2 hours during each cycle at normal voltage and deenergized for 1/2 hour.

* 4.4.3.3.2 Performance characteristics to be measured - The performance characteristics to be measured shall be as specified in 6.9.

* 4.4.3.3.3 Failure criteria - In addition to the requirements of MIL-STD-781, the following requirements shall be used to determine when a failure has occurred during the test:

(a) Whenever performance characteristics fall below the acceptance requirement (4.4.3.3.2) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other to fail.

* 4.4.3.3.4 Preventive maintenance - During the periods of the test, no preventive maintenance, other than simple adjustments normally performed on the adapter by the operator, shall be permitted. However, a log of all such adjustments shall be kept. Excessive adjustments may be considered as failures by the procuring activity.

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* 4.4.4 Special tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

* 4.4.4.1 Special test schedule - Selection of equipments for special tests shall be made as follows:

(a) On an early equipment after an engineering or material change.

(b) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity).

* 4.4.4.2 Scope of tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

* 4.4.5 Equipment failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(a) Determine the cause of failure.

(b) Determine if the failure is an isolated case or design defect.

(c) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(d) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

* 4.5 Longevity tests - Longevity tests shall be performed in accordance with Test Plan XXVIII of MIL-STD-781. The longevity test procedures shall be similar to those of 4.4.3. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for Longevity Tests.) When reliability tests are conducted, the longevity tests may be omitted if, during the reliability tests, two or more equipments receive at least 2000 hours each of test time.

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* 4.5.1 Reconditioning of longevity test samples - An equipment which has been subjected to the longevity test shall be reconditioned as follows:

(a) On completion of the longevity test, the equipment may be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(b) After reworking, the contractor shall resubmit the equipment for acceptance.

* 4.6 Test procedures - The procedures used for conducting pre-production tests, acceptance tests and longevity tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Test conditions -

4.7.1 Standard conditions - Unless otherwise specified, all tests required by this specification shall be made under the following conditions:

Temperature	Room ambient 25 ±5° C
Pressure	Normal atmospheric (approximately 29.92 inches Hg.)
Humidity	Room ambient up to 90% relative humidity

4.7.2 Test voltage - Unless otherwise specified, all tests shall be conducted with a power supply of 115 ±1 volts, 400 ±5 cycles alternating current and 28 ±1 volt (grounded negative) direct current.

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4.7.3 Test control transformer - The test control transformer shall be a synchro with the following nominal electrical characteristics:

Input Voltage	11.8 Volts
Input Impedance (Rotor open circuited)	20 + j 100 Ohms
Rotor Resistance	42 Ohms
Stator Resistance	10.8 Ohms
Output Voltage	22.4 Volts

It shall have a scale from 0 to 360 degrees readable to three minutes. The following procedure shall be used to calibrate the control transformer.

4.7.3.1 Designate at will and label the leads of the rotor "Hot" and "Cold."

4.7.3.2 Select one of the stator leads, designate and label it "Z" and tie this lead to the "Cold" rotor lead.

4.7.3.3 Apply 26 V, 400 cycles AC to the "Hot" lead and ground the "Cold" and "Z" leads. Rotate the rotor or stator until the output of the remaining leads are at a "null" or minimum voltage, as indicated on a vacuum tube voltmeter.

4.7.3.4 Measure the voltage between the "Hot" lead and either of the remaining stator leads. If this voltage is less than the excitation voltage of the rotor, the synchro rotor is at "Low" null or the 270-degree position. If the voltage is greater than the rotor excitation voltage, the synchro rotor is at "High" null or the 90-degree position.

4.7.3.5 The rotor shall be placed at the "High" null or 90-degree position. A voltmeter shall then be connected across one of the remaining stator leads and the ground, with the second lead open. When the synchro rotor is turned clockwise, the voltage will increase before decreasing if the "X" stator lead is connected. All the leads of the synchro shall be labeled in accordance with the results of the above test.

4.7.3.6 In use, the test control transformer may be positioned by hand or by a servo controlled motor or equivalent means.

* 4.7.4 Test equipment - The adapter tests shall be made with the proper MA-1 Compass System conforming to MIL-C-17858 or approved simulated components which duplicate all the necessary functions of the MA-1 Compass System.

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A type ML-1 Compass Transmitter conforming to MIL-T-19576 shall be used as the accessory transmitter to the MA-1 Compass System.

4.7.5 Field strength - Unless otherwise specified, the compass transmitter shall be located in a standard magnetic field that is undirectional and unvarying such as the earth's magnetic field having a horizontal field strength of 0.18 ±0.02 oersted and a vertical field strength of 0.54 ±0.02 oersted.

4.7.6 Test reading vibration - All performance tests for which readings of the compass system are required shall be conducted with the directional gyro subjected to a vibration of 0.002 to 0.005 inch amplitude and a frequency of 25 to 33 cycles per second along a vertical axis.

4.7.7 Measurements - All measurements shall be made with instruments whose accuracy has been verified. If tests are conducted at the contractor's plant, the accuracy of the instruments and test equipment shall be verified periodically by the contractor to the satisfaction of the procuring activity.

4.8 Test methods -

4.8.1 Examination of product - Each adapter shall be examined externally to determine conformance with the applicable drawings and with all the requirements of this specification not covered by specific tests.

4.8.2 Dielectric strength - A potential of 550 volts (root mean square) alternating current at commercial frequency shall be applied between isolated pins, and between pins and the case for a period of 5 seconds. That potential shall not be applied to direct current, grounding terminals or synchro terminals. Synchro terminals shall be tested at a potential of 375 volts (rms). There shall be no breakdown of insulation or any other permanent damage to the instrument as a result of this test.

* 4.8.3 Scale error - The adapter shall be connected to a Type MA-1 Compass System or approved simulated components. The angular difference between the signal input to the adapter and the signal output from each of the transmitting synchros shall be determined at each of the 24 equally spaced headings including the cardinal headings. The total errors shall not exceed ±1/4 degree.

4.8.4 Leakage - Each internal hermetically sealed subassembly case of the adapter shall be tested for case leakage with a mass spectrometer or an equivalent, at least equally sensitive, quantitative leak test. The rate of leakage of the gas or mixture of gasses in the case, adjusted to a pressure differential of one atmosphere, shall not exceed one micron cubic foot per hour (10^{-5} cubic centimeters per second).

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* 4.8.5 Compensator range - The adapter shall be connected to the Type MA-1 System. The adapter compensation shall be set to zero. The servo input to the adapter shall be set to North heading and the compass transmitter adjusted so that the system is at equilibrium. The North compensator adjustment shall be set to give full positive compensation. The compass transmitter shall be reset so that the system is at equilibrium with the servo input to the adapter at North heading. The change in compass heading shall be not less than 3 degrees nor more than 7 degrees. The North compensator adjustment shall then be set to give full negative compensation and the compass transmitter reset to bring the system into equilibrium with the servo input to the adapter at North heading. The change in compass heading from the heading with zero compensation shall be not less than 3 degrees nor more than 7 degrees. The procedure above shall be repeated at each successive 15° point until all 24 points have been tested.

4.8.6 Servo response - The transmitting synchro outputs shall follow smoothly a rotation input with less than 0.2 mechanical degree error per revolution per minute up to 7 rpm with a maximum error of 1.4 mechanical degrees. It shall also follow smoothly a sine wave input of one cycle per second and 10 degrees peak to peak amplitude with less than 20 degrees phase lag.

4.8.7 Power consumption - The adapter shall be placed in the "mode" of compass system operation that consumes the most power within the requirements of MIL-STD-704. The rated voltages and frequencies of power shall be applied. The AC power shall not exceed 40 VA and the DC power shall not exceed 15 watts.

4.8.8 Voltage and frequency variation - At any combination of voltage and frequency, the adapter shall meet the requirements of the Compensated Scale Error Test (4.8.11). The varying voltages and frequencies shall be in accordance with the requirements of MIL-STD-704.

* 4.8.9 Low temperature operation - The adapter shall be tested in accordance with Step 2, Temperature and Altitude Test of MIL-T-5422 for 4 hours. During the specified temperature, the adapter shall be subjected to and meet the requirements of the Compensated Scale Error Test (4.8.11).

* 4.8.10 High temperature operation - The adapter shall be tested in accordance with Step 6, Temperature and Altitude Test of MIL-T-5422. During the specified temperature, the adapter shall be subjected to and meet the requirements of the Compensated Scale Error Test (4.8.11).

* 4.8.11 Compensated scale error - Connect the adapter to a complete Type MA-1 Compass System (MIL-C-17858). Select any synchro in the Compass System and any synchro in the adapter as the output references. Remove all compensation in the adapter. The output references shall be set at 0° and

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the compass transmitter of the compass system set to bring the system into an equilibrium condition. Record the precise compass directional heading. Repeat this at each required heading over the full 360 degrees. Set the compass transmitter at a heading determined by the algebraic sum of the 0° heading recorded above and 3 Sin 0°. Restore the adapter compensation and adjust the 0° tap until the system is in equilibrium with the output reference heading set at 0°. Now set the compass transmitter at a heading equal to the algebraic sum of the 15° uncompensated heading recorded above and 3 Sin 15°. Adjust the 15° compensation tap until the system is in equilibrium with the output reference heading set at 15°. Repeat the above procedure at each compensation setting heading. After compensation, the compensated scale error test shall be conducted at each of the required output reference headings. The output reference headings shall consist of those listed in Table II as a minimum requirement. The compass transmitter shall be set to bring the compass system into equilibrium at each of the required output reference headings. The precise directional heading of the compass transmitter shall be recorded for each heading. The compensated scale error is the difference between the algebraic sum of the output reference heading plus 3 Sin (output reference heading) and the recorded compass transmitter directional heading. When subsequent Compensated Scale Error Tests are required, the compensation taps shall not be readjusted and the test shall consist of setting the output reference heading and compass transmitter to the headings required to determine the compensated scale error. When subsequent compensated scale error tests are conducted, including those at adverse environments, the total maximum scale error shall not diverge from the initial scale error by more than 0.5° and in no case shall the total maximum scale error exceed 0.7°.

* 4.8.12 Vibration - The adapter shall be tested in accordance with Parts I and II, Vibration Tests of MIL-T-5422. At the conclusion of the test procedure, the adapter shall be subjected to and shall meet the requirements of the Individual Tests.

* 4.8.13 Radio noise interference - The adapter properly connected, shall be tested for conducted and radiated radio noise interference in accordance with MIL-I-6181.

* 4.8.14 Temperature-altitude - The adapter shall be tested in accordance with the Temperature and Altitude Tests of MIL-T-5422. At the conclusion of the test procedure, the adapter shall be subjected to and shall meet the requirements of the Individual Tests.

* 4.8.15 Temperature shock - The adapter shall be subjected to the Temperature Shock Test, Procedure I of MIL-T-5422. At the conclusion of the test procedure, the adapter shall be subjected to and shall meet the requirements of the Individual Tests.

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TABLE II

Output Reference Heading	Compass Correction	Output Reference Heading	Compass Correction
* 0	0	*180	0
7.5	0.39	187.5	-0.39
* 15	0.78	*195	-0.78
22.5	1.15	202.5	-1.15
* 30	1.50	*210	-1.50
37.5	1.83	217.5	-1.83
* 45	2.12	*225	-2.12
52.5	2.38	232.5	-2.38
* 60	2.60	*240	-2.60
67.5	2.77	247.5	-2.77
* 75	2.90	*255	-2.90
82.5	2.97	262.5	-2.97
* 90	3.00	*270	-3.00
97.5	2.97	277.5	-2.97
*105	2.90	*285	-2.90
112.5	2.77	292.5	-2.77
*120	2.60	*300	-2.60
127.5	2.38	307.5	-2.38
*135	2.12	*315	-2.12
142.5	1.83	322.5	-1.83

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TABLE II (Continued)

Output Reference Heading	Compass Correction	Output Reference Heading	Compass Correction
*150	1.50	*330	-1.50
157.5	1.15	337.5	-1.15
*165	0.78	*345	-0.78
172.5	0.39	352.5	-0.39

*Compensation Setting Headings

* 4.8.16 Impact shock - The adapter shall be tested in accordance with the Shock Test of MIL-T-5422. At the conclusion of the test procedure, the adapter shall be subjected to and shall meet the requirements of the Individual Tests.

* 4.8.17 Humidity - The adapter shall be tested in accordance with the Humidity Test of MIL-T-5422 except 71° C shall be substituted for 50° C in Steps 1 and 2. At the conclusion of the test procedure, the adapter shall be subjected to and shall meet the requirements of the Individual Tests. Examination for deterioration shall be made between 24 and 48 hours after completion of the humidity exposure. There shall be no evidence of corrosion or other deterioration which will affect subsequent operation of the adapter.

* 4.8.18 Salt spray - The adapter shall be tested in accordance with the Salt Spray Test of MIL-T-5422, except that the test period shall be 50 hours. At the conclusion of the test procedure, the adapter shall be subjected to and shall meet the requirements of the Individual Tests.

* 4.8.19 Fungus resistance - The adapter shall be tested in accordance with the Fungus Resistance Test of MIL-T-5422 except that the test period shall be 14 days. At the conclusion of the test procedure, any evidence of the growth of fungi found upon inspection shall be cause for rejection of the adapter or of any lot represented by the tested adapter and the adapter shall be subjected to and shall meet the requirements of the Individual Tests.

* 4.9 Reconditioning of tested equipment - Equipment which has been subjected to acceptance and longevity tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

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* 4.10 Presubmission testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

* 4.11 Rejection and retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

5. PREPARATION FOR DELIVERY

* 5.1 Packaging - The compensating adapter shall be packaged in accordance with MIL-STD-794, Level A. The method of preservation shall be in accordance with MIL-P-116, Method IC, utilizing a rectangular container. The container shall conform to MIL-C-4150.

* 5.2 Packing - The adapter shall be packed in accordance with MIL-STD-794, Levels A, B, or C as specified in the contract or order (see 6.6). The package and exterior shipping container may be one and the same container.

* 5.3 Rough handling tests - Rough handling tests shall apply to all levels of packaging and packing. The tests performed on a prototype example shall be Rough Handling, Dummy Load and Preproduction Tests of MIL-STD-794. No vibration tests are required. Two pilot shipping containers are required to be tested. The heights of drop shall be 30 inches for the free-fall flat drop and 36 inches for the free-fall cornerwise drop. All testing shall be done on all axes and in each direction in each axis and with actual equipment as packaged for shipment. After tests, the equipments shall not suffer damage or subsequently fail to provide performance as specified. All testing shall be done at the Contractor's plant. The contractor shall notify the procuring activity in sufficient time in advance of prototype testing to permit Government representation at the test site.

5.4 Marking - The interior and exterior containers shall be marked as specified in MIL-STD-129 and as follows:

* 5.4.1 Precautionary marking - The following precautionary markings warning of the fragility of the item shall appear on two opposite sides of each interior package and shipping container. Containers shall be marked with arrows on two opposite sides of each interior package and shipping container to indicate which surface must be always up:

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FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE
UP



In addition, containers shall be marked "Reusable, OPEN WITH CARE" in letters at least 2 inches high, and on two opposite sides of each container.

6. NOTES

6.1 Intended use - The adapter covered by this specification is intended for use in fighter, attack, patrol, transport and cargo type aircraft whose range of operation would be worldwide. The adapter will provide deviation compensation for the Type MA-1 Compass System and additional azimuth references in the form of synchro signals to other control and navigational systems.

* 6.2 Test values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ±0.5 VDC and 115 ±1.0 VAC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptance standard of performance.

* 6.3 Performance objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

* 6.4 As a general rule nonrepairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the nonrepairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

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* 6.5 Precedence of documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specification.

(b) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

* 6.6 Ordering data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(c) The quantity and MS part number of the instrument desired.

* 6.7 Asterisk - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.8 Items of equipment - The adapters covered by this specification shall be of the following types:

Part Number	No. of Outputs	Associated Compass
MS28107-1	8 - 4 Power transmitter outputs	MA-1 Compass (System A)

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Part Number	No. of Outputs	Associated Compass
MS28107-2	3 - 4 Power transmitter outputs	MA-1 Compass (System A)
MS28107-3	2 - 4 Power transmitter outputs and 1 - P-1 Automatic Pilot clutched synchro output	MA-1 Compass (System A)
MS28107-4	8 - 4 Power transmitter outputs	MA-1 Compass (System B)
MS28107-5	3 - 4 Power transmitter outputs	MA-1 Compass (System B)
MS28107-6	3 - 4 Power transmitter outputs and 1 - P-1 Automatic Pilot clutched synchro output	MA-1 Compass (System B)
MS28107-7	3 - 4 Power transmitter outputs and 1 - P-1 Automatic Pilot clutched synchro output	MA-1 Compass (System A)

* 6.9 Performance characteristics to be measured - At least once each week, the compensating adapter shall be subjected to and shall meet the requirements of the following tests:

(a) Scale Error

(b) Compensator Range

(c) Compensated Scale Error

A daily observation (on each workday) shall be made of each adapter in order to detect obvious failures.

6.10 Definitions -

6.10.1 Hermetic seal - A hermetic seal shall be a perfectly closed and airtight seal made between metallic and metallic or between metallic and vitric materials. A hermetic seal is not intended to include seals which are dependent upon gaskets.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-A-22424A(AS) ADAPTERS, COMPENSATING, MS28107, MA-1 COMPASS SYSTEM

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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